

A decorative graphic on the left side of the slide consisting of overlapping geometric shapes. It includes a blue parallelogram, a light green parallelogram, and a dark grey parallelogram, all oriented diagonally.

Credit Card Approval: Improving Model with Diverse Machine Learning Techniques



We are Team of 3!

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Agenda:

1.	Introduction
2.	Problem statement
3.	Related Works
4.	Data
5.	Methodology
6.	Results
7.	Conclusion



Introduction

- Credit card fraud poses a major risk to financial institutions and customers.
- Detecting fraudulent transactions is challenging due to highly imbalanced data (only 0.17% fraud).
- Traditional accuracy metrics fail to capture true performance in such settings.
- By applying advanced preprocessing, imbalance handling, and model tuning, we aim to create a system that improves reliability and reduces financial risk.



Problem statement

- Credit approval & fraud detection are critical in modern financial systems.
- Lenders must assess applications accurately to avoid:
 - Financial loss
 - Unfair decisions & bias
- Goal: Increase reliability & reduce fraud risk by improving model performance to ~97–98%.



Why this matters

- Fraud costs financial institutions billions per year.
- Traditional accuracy is misleading due to extreme class imbalance.
- Missing a fraud case (false negative) has high cost.
- Need a system that:
 - Detects rare fraud cases
 - Minimizes false alarms
 - Works reliably in real-world banking pipelines



Related Works

- Fraud detection is widely studied as an imbalanced classification problem.
- Prior research highlights challenges such as extreme imbalance, evolving fraud patterns, and real-time detection needs.
- Survey papers (Lucas & Jurgovsky, 2020; Dastidar et al., 2024) compare machine learning methods including:
 - Logistic Regression
 - Decision Trees
 - Random Forests
 - Gradient Boosting
 - Neural Networks
- Studies show tree-based ensemble methods (especially Random Forests) consistently achieve strong performance on imbalanced credit card datasets.
- Research also emphasizes using PR-AUC, ROC-AUC, and F1-score instead of raw accuracy for reliable evaluation.



Data

Credit Card Fraud Detection Dataset (Kaggle)

- 284,807 transaction rows
- 30 features
 - PCA-transformed V1-V28
 - Time, Amount
- Fraud cases: 492 (~0.17%)
- Highly imbalanced → requires special evaluation & model handling

```
Class counts:
  Class
0      284315
1         492
Name: count, dtype: int64

Class distribution (%):
  Class
0      99.827251
1       0.172749
Name: count, dtype: float64
```





Methodology

To improve performance, we built a complete ML pipeline:

1. Data Preprocessing
 - a. Standardization of Amount
 - b. Stratified train/test split
 - c. No missing values
2. Imbalance Handling
 - a. Class weighting
 - b. SMOTE during training



Methodology cont.

3. Modeling

- Baseline Logistic Regression
- Tuned Random Forest

4. Evaluation

- PR-AUC, ROC-AUC, F1 (fraud)
- Confusion matrix

5. Hyperparameter Tuning

- 3-fold cross-validation
- RandomizedSearchCV

Baseline Model

- Logistic Regression
 - Strong accuracy
 - Weak recall for fraud class
 - Motivated use of tree-based methods for non-linear interactions

Confusion Matrix:

```
[[56851  13]  
 [  35  63]]
```

Classification Report:

	precision	recall	f1-score	support
0	1.00	1.00	1.00	56864
1	0.83	0.64	0.72	98
accuracy			1.00	56962
macro avg	0.91	0.82	0.86	56962
weighted avg	1.00	1.00	1.00	56962

ROC-AUC: 0.9595894034316034



Random Forest

- Why Random Forest?
 - Handles non-linear patterns
 - Naturally robust to noise
 - Works well on imbalanced tabular data
 - Supports class weighting
- Key settings:
 - `n_estimators = 200`
 - `max_depth = None`
 - `min_samples_split = 5`
 - `min_samples_leaf = 1`
 - `SMOTE + class_weight = "balanced"`



Hyperparameter Tuning

Used GridSearchCV with PR-AUC as scoring metric.

Best parameters:

- 200 trees
- min samples split = 5
- leaf = 1
- no max depth

Result: Best validation PR-AUC $\approx$ 0.844

```
Fitting 3 folds for each of 10 candidates, totalling 30 fits
Best params: {'model__n_estimators': 200, 'model__min_samples_split': 5, 'model__min_samples_leaf': 1,
              'model__max_depth': None}
Best PR-AUC (cv): 0.844026112515706
```

Test Set Results

Final Model Performance:

- Accuracy: 99.95%
- ROC-AUC: 0.9831
- PR-AUC: 0.8734
- Fraud F1-score: 0.8377
- Fraud Precision: 0.8602
- Fraud Recall: 0.8163

These results confirm the model detects most fraud cases while minimizing false positives.

Confusion Matrix:

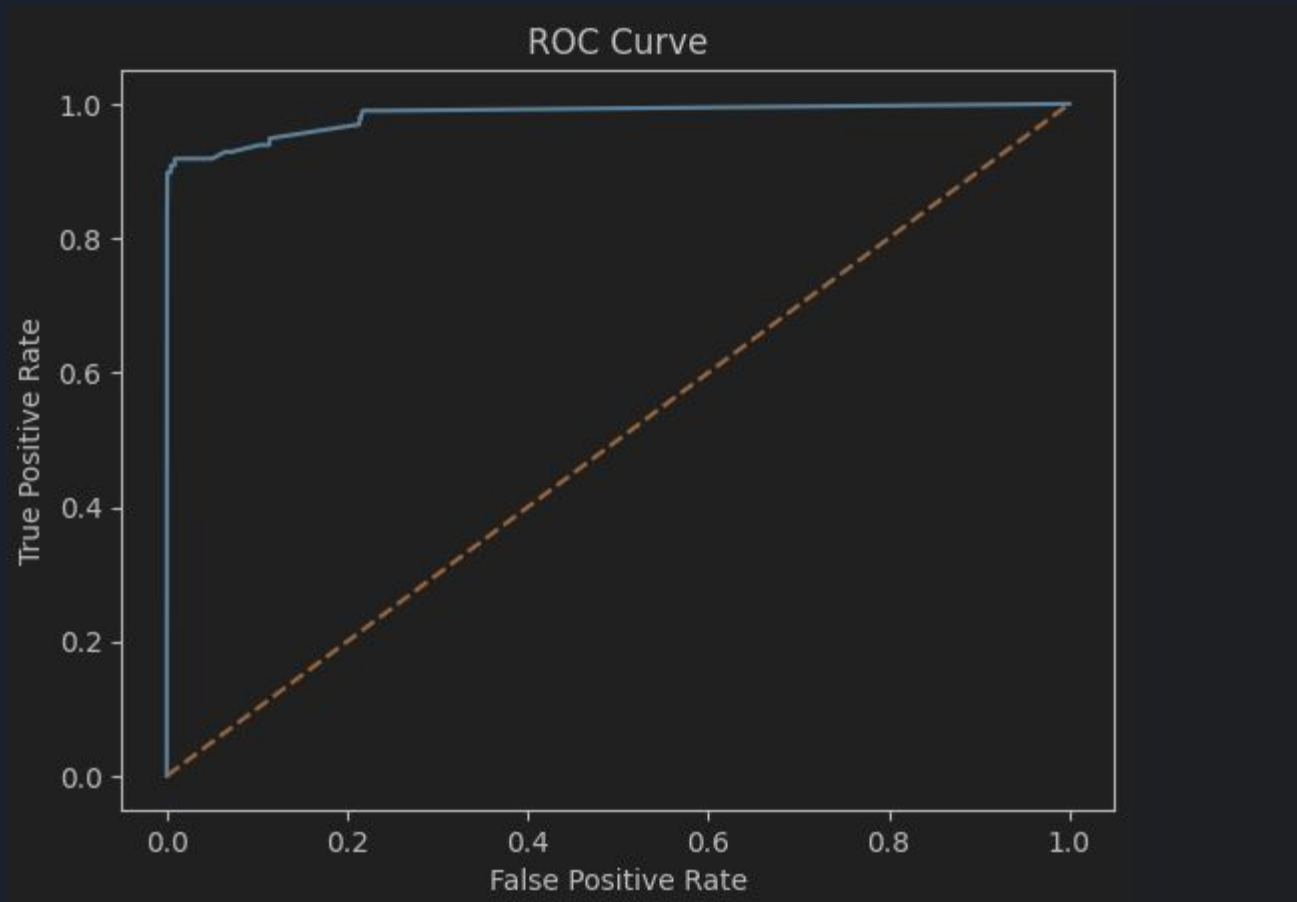
- Only 13 legitimate transactions incorrectly flagged
- 81.6% of fraud detected
- Very low false-positive rate

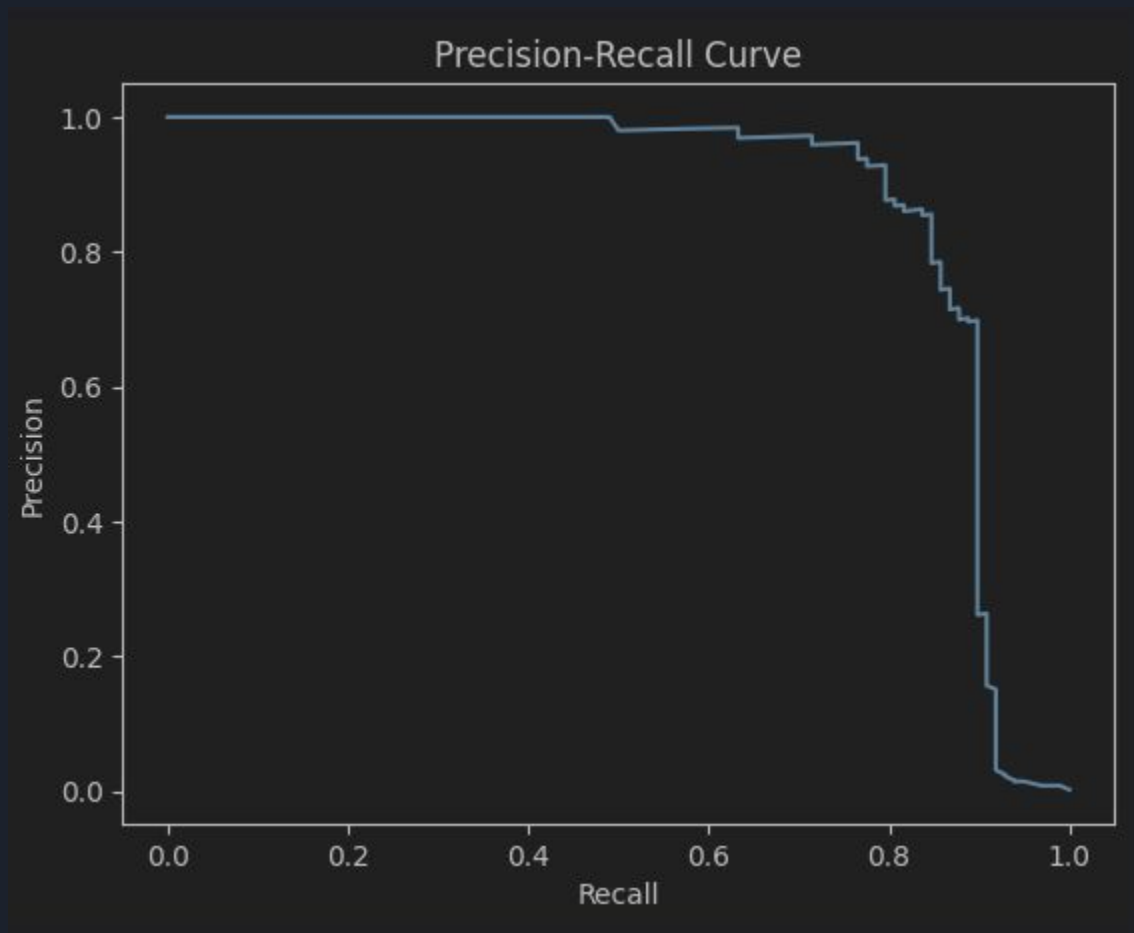
Classification Report:

	precision	recall	f1-score	support
0	0.9997	0.9998	0.9997	56864
1	0.8602	0.8163	0.8377	98
accuracy			0.9995	56962
macro avg	0.9299	0.9080	0.9187	56962
weighted avg	0.9994	0.9995	0.9994	56962
ROC-AUC:	0.9831			
PR-AUC:	0.8734			
F1-score (threshold=0.5):	0.8377			

Confusion Matrix:

```
[[56851  13]
 [  18  80]]
```







Limitations

- Dataset is from 2013—may not reflect modern fraud patterns
- PCA-transformed features → lower interpretability
- SMOTE may introduce synthetic noise
- Fraud patterns evolve → model must be retrained continuously



Future Work

- Use cost-sensitive learning for real-money risk tuning
- Deploy model with monitoring (MLOps)
- Try Deep learning models
- Add explainability tools (SHAP) for analysts
- Adjust decision thresholds for business needs



Conclusion

- Built full ML pipeline for fraud detection
- Tuned Random Forest achieved strong performance
- High detection rate, low false positives
- Demonstrated that classical ML can be highly effective on imbalanced fraud datasets